

**Aim:**

To write and execute a Java program to check whether the given number is both a **prime number and a palindrome**

**Algorithm: Prime Palindrome**

**Step 1:** Start the program.

**Step 2:** Import Scanner for console input.

**Step 3:** Define the isPrime() method.

**Step 4:** Check if the number is less than 2. If yes, return false.

**Step 5:** Check divisibility from 2 to  $n/2$ . If divisible, return false; otherwise return true.

**Step 6:** Define the isPalindrome() method.

**Step 7:** Store the original number and reverse the number using the digits.

**Step 8:** Compare the original number with the reversed number. If equal, return true; otherwise return false.

**Step 9:** In main(), read the number from the user.

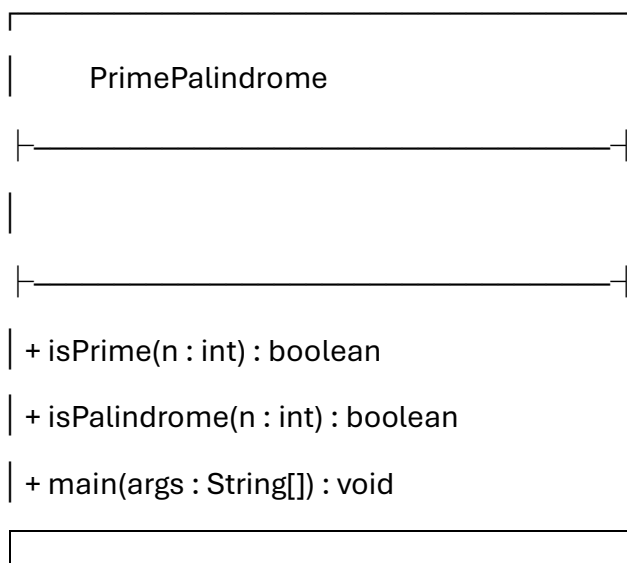
**Step 10:** Call isPrime() and isPalindrome().

**Step 11:** If both return true, display "**Prime Palindrome number**".

**Step 12:** Otherwise, display "**Not a Prime Palindrome number**".

**Step 13:** Close the Scanner.

**Step 14:** Stop the program.

**UML Diagram:**

Code:

```
import java.util.Scanner;
```

```
public class PrimePalindrome {
```

```
    // Method to check prime
```

```
    static boolean isPrime(int n) {
```

```
        if (n < 2)
```

```
            return false;
```

```
        for (int i = 2; i <= n / 2; i++) {
```

```
            if (n % i == 0)
```

```
                return false;
```

```
        }
```

```
        return true;
```

```
    }
```

```
    // Method to check palindrome
```

```
    static boolean isPalindrome(int n) {
```

```
        int original = n;
```

```
        int reverse = 0;
```

```
        while (n > 0) {
```

```
            int digit = n % 10;
```

```
            reverse = reverse * 10 + digit;
```

```
            n = n / 10;
```

```
        }
```

```

        return original == reverse;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

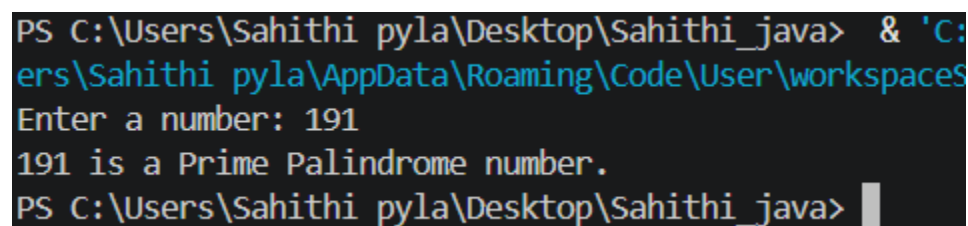
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (isPrime(n) && isPalindrome(n))
            System.out.println(n + " is a Prime Palindrome number.");
        else
            System.out.println(n + " is not a Prime Palindrome number.");

        sc.close();
    }
}

```

Output:



```

PS C:\Users\Sahithi pyla\Desktop\Sahithi_java> & 'C:\Users\Sahithi pyla\AppData\Roaming\Code\User\workspaces\Sahithi pyla\src\main\java\com\sahithi\palindrome\PrimePalindrome.java'
Enter a number: 191
191 is a Prime Palindrome number.
PS C:\Users\Sahithi pyla\Desktop\Sahithi_java>

```